

Department of Electrical Engineering
Indian Institute of Technology Kharagpur
Machine Learning for Signal Processing Laboratory
(EE69210)
Spring, 2023-24

Experiment 9: Kernel Density Estimation

Grading Rubric

	Tick the best applicable per row			Points
	Below Expecta- tion	Lacking in Some	Meets all Expecta- tion	
Completeness of the report				
Organization of the report (5 pts)				
Quality of figures (5 pts)				
Dataset creation (10 pts)				
Histogram plots and density map- pings. (25 pts)				
Plot showing how changing the bins can morph the modality of data from unimodal to bimodal, etc. (20 pts)				
Visual representation of the data distribution and the density around each point in the data. (25 pts)				
Smoothened KDE. (10 pts)				
TOTAL (100 pts)				

1 Code of Conduct

Students are expected to behave ethically both in and out of the lab. Unethical behaviour includes, but is not limited to, the following:

- Possession of another person's laboratory solutions from the current or previous years.
- Reference to, or use of another person's laboratory solutions from the current or previous years.
- Submission of work that is not done by your laboratory group.
- Allowing another person to copy your laboratory solutions or work.
- Cheating on quizzes.

The rules of laboratory ethics are designed to facilitate these goals. We emphasize that laboratory TAs are available to help the student understand the basic concepts and answer the questions asked in the laboratory exercises. By performing the laboratories independently, students will likely learn more and improve their performance in the course as a whole. Please note that it is the student's responsibility to ensure that the content of their graded laboratories is not distributed to other students. If there is any question as to whether a given action might be considered unethical, please see the professor or the TA before you engage in such actions.

2 Kernel Density Estimation

Kernel Density Estimation (KDE) serves as an algorithm that pushes the boundaries of the mixture-of-Gaussians concept to its logical extreme. Unlike methods with a predetermined number of Gaussian components, KDE takes a decidedly non-parametric approach by employing a mixture that includes one Gaussian component for every individual data point. This unique characteristic transforms KDE into an exceptionally versatile and flexible density estimator.

In the upcoming section, our journey delves into the intrinsic motivation behind the development of KDE, unraveling the rationale that propels its adoption as a non-parametric methodology. By embracing the principle of assigning a Gaussian component to each data point, KDE transcends the limitations of pre-defined parameters, offering a nuanced and adaptable means of estimating probability density functions.

Our exploration extends beyond the theoretical underpinnings of KDE, as we venture into its practical applications. Through this comprehensive examination, we aim to provide a thorough understanding of the method's utility across various domains. By doing so, we lay a robust foundation for readers to grasp the intricacies of KDE, empowering them to leverage this non-parametric density estimation technique effectively in their analytical pursuits.

2.1 Histograms of the data generated

The density estimator is an algorithm that seeks to model the probability distribution that generates a dataset. For one-dimensional data, you are probably already familiar with one simple density estimator: the histogram. A histogram divides the data into discrete bins, counts the number of points that fall in each bin, and then intuitively visualizes the results. We use the normalized values from the histogram to get the output as probabilities instead of the input distribution.

2.2 Issues with histograms as a density estimator

One of the issues with using a histogram as a density estimator is that the choice of bin size and location can lead to representations that have qualitatively different features. For example, if we look at a version of this data with only 20 points, the choice of how to draw the bins can lead to an entirely different interpretation of the data.

2.3 Creating a smoother histogram

Taking a step back, let's consider a histogram as a collection of blocks, each one placed within a designated bin and stacked atop corresponding points in the dataset. The issue with our two binning methods arises from the fact that the height of this block stack often reflects not the actual density of nearby points but rather coincidences in how the bins align with the data points. This misalignment is a potential cause for the suboptimal histogram results observed.

However, what if, instead of stacking the blocks in alignment with the bins, we were to stack the blocks in alignment with the points they represent? While this may result in blocks that aren't perfectly aligned, we can overcome this by aggregating their contributions at each location along the x-axis to derive the final result. This alternative approach ensures that the representation is not skewed by arbitrary bin placements, offering a more accurate depiction of the underlying data density.

This smoothed-out plot, with a Gaussian distribution, contributed at the location of each input point, gives a much more accurate idea of the shape of the data distribution, and one which has much less variance (i.e., changes much less in response to differences in sampling).

3 Experiments on Kernel Density Estimation

3.1 Importing essential libraries

```
1 #importing libraries
2 %matplotlib inline
3 import matplotlib.pyplot as plt
4 import seaborn as sns; sns.set()
5 import numpy as np
```

Listing 1: Code to import essential libraries

3.1.1 Creating the dataset.

```
1 def make_data(N, f=0.3, rseed=1):
2     rand = np.random.RandomState(rseed)
3     x = rand.randn(N)
4     x[int(f * N):] += 5
5     return x
6
7 x = make_data(1000)
```

Listing 2: Code to create the dataset

3.1.2 Assignments to solve and report

Write the codes for the following tasks in a .ipynb file, including all visualizations, and submit the executed file. Submit also a separate pdf of your report on this experiment. This report should describe your observations and reasoning while executing these experiments.

1. Rerun all the steps discussed till now, rerunning all the codes. Check your consistency in generating data similar to those presented earlier.

2. Plot the histogram of density mappings.
3. Create 20 random data points using the above function. Create 10 equally spaced bins between -5 and 10. Add an offset of 0 and 0.6 to the bins and plot the histogram of the density mappings of the data. See how the data distribution changes now.
4. Represent the histogram by using vertical lines and rectangles to illustrate the frequency distribution of data points within specified bins along the x-axis
5. Generate a plot that combines a kernel density estimate with vertical lines representing individual data points. This provides a visual representation of the data distribution and the density around each point in the dataset.
6. Use Gaussian kernels from the normal distribution to compute a kernel density estimate and visualize it along with individual data points through filled areas and vertical lines, respectively.